

SUPERMARKET SALES ANALYSIS REPORT

Executive Summary

This project focuses on analyzing supermarket sales data to understand customer purchasing behavior, product performance, sales trends, and payment preferences. The objective was to identify key revenue drivers and generate actionable business recommendations that could improve sales performance and customer satisfaction.

Dataset Overview

The dataset contains transaction-level sales information including product categories, customer purchases, payment methods, and total sales values. The data was analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn.

Methodology

The following steps were performed:

1. Data loading and inspection
2. Missing value and duplicate record checks
3. Exploratory Data Analysis (EDA)
4. Statistical analysis
5. Data visualization
6. Business insight generation

Key Findings

- Product line performance varied significantly across categories.
- Certain product categories generated a higher percentage of total revenue.
- Customer spending patterns showed clear preferences for specific products.
- Payment methods influenced purchasing convenience and transaction frequency.
- Sales distribution differed across locations and customer groups.

Visualizations Generated

- Revenue by Product Line
- City-wise Revenue Distribution
- Payment Method Analysis

Business Insights

1. High-performing product categories contribute a significant share of total sales.
2. Customer purchasing behavior is influenced by convenience and product availability.
3. Certain cities generate consistently higher revenue.
4. Electronic payment methods are increasingly preferred.
5. Product demand trends can be used for inventory planning.

Recommendations

- Increase inventory for high-demand products.
- Implement targeted marketing campaigns.
- Offer promotional discounts during peak purchasing periods.
- Improve stock forecasting based on demand patterns.
- Expand digital payment options.

Conclusion

The supermarket sales analysis successfully identified important revenue drivers and customer behavior trends. The findings can support better inventory management, targeted marketing strategies, and improved business decision-making.